THE ELEMENT-ORDER COLLISION PROBABILITY DISTINGUISHES THE 47 GROUPS OF ORDER 120

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Abstract. For a finite group G, let $N_k(G)$ denote the number of elements of order k in G, and define the *element-order collision probability*

$$P_2(G) \;=\; \sum_{k\geq 1} \left(\frac{N_k(G)}{|G|}\right)^2 .$$

This is the Rényi-2 collision entropy of the order-type distribution viewed as a probability mass function on $\mathbb{Z}_{>0}$. We show by direct enumeration over the GAP `SmallGroups` library that the 47 pairwise non-isomorphic groups of order 120 have 47 pairwise distinct values of P_2, so P_2 is a complete invariant on this order. Among these 47 groups, 27 ordered pairs (G_1, G_2) satisfy the *single-bucket-swap* property: their order-type distributions differ in exactly two orders k_1, k_2 by the same count $n \geq 1$, equivalently, passing from G_1 to G_2 redistributes n elements from order k_1 to order k_2. The pair $(S_5, \mathrm{SL}(2,5))$ realises this pattern with $n = 24$, $k_1 = 2$, $k_2 = 10$, giving the explicit gap $P_2(S_5) - P_2(\mathrm{SL}(2,5)) = 1/300$. All claims are reproducible from a self-contained GAP script of fewer than 50 lines.

1. Introduction

The multiset of element orders of a finite group G, often called the *order type* of G [6], has been intensively studied as a coarse but information-rich invariant. Thompson asked whether sameness of order type forces sameness of solvability; the question was answered negatively by Piwek [5] in 2024. Cameron and Dey [2] place the order type in a partial order and study its combinatorial structure. The sum of element orders $\psi(G) = \sum_{x\in G} |x|$ is a scalar derived from the order-type distribution and has accumulated a substantial literature since Amiri, Jafarian Amiri and Isaacs [1]; see also [4] and references therein.

Less attention has been paid to the second moment of the order-type distribution, equivalently the *collision probability*

$$P_2(G) \;=\; \Pr_{(x,y)\in G^2}[|x| = |y|] \;=\; \sum_{k\geq 1} \left(\frac{N_k(G)}{|G|}\right)^2 ,$$

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where the probability is over two uniformly independent draws from G and $|x|$ denotes the order of x. This is the standard collision probability associated with the order-type distribution, equivalently the Rényi entropy of order 2,

$$H_2\big(\mathrm{order}(G)\big) \;=\; -\log_2 P_2(G).$$

The aim of this note is to record a small but striking computational fact: on the order $n = 120$, the scalar P_2 is sharp enough to distinguish all 47 pairwise non-isomorphic groups, and the pairs of groups whose P_2 values differ by a particularly small rational gap admit a clean structural description.

2. Definitions and notation

Let G be a finite group with $|G| = n$. For $k \in \mathbb{Z}_{>0}$ write

$$N_k(G) \;=\; \#\{x \in G : |x| = k\}.$$

Then $\sum_k N_k(G) = n$ and the *order-type* of G is the multiset $\{|x| : x \in G\}$, equivalently the function $k \mapsto N_k(G)$.

Definition 1. The *element-order collision probability* of G is

$$P_2(G) \;=\; \sum_{k\geq 1} \left(\frac{N_k(G)}{|G|}\right)^2 \;=\; \frac{1}{|G|^2}\sum_k N_k(G)^2.$$

Equivalently $P_2(G)$ is the probability that two uniformly random elements of G have the same order. The denominator $|G|^2$ implies that all P_2 values for groups of order n are rationals with denominator dividing n^2.

Definition 2. Two finite groups G_1, G_2 with $|G_1| = |G_2|$ form a *single-bucket-swap pair* with parameters (k_1, k_2, n) if the difference of their order-type distributions is supported on exactly $\{k_1, k_2\}$ with

$$N_{k_1}(G_2) - N_{k_1}(G_1) \;=\; -n \quad \text{and} \quad N_{k_2}(G_2) - N_{k_2}(G_1) \;=\; +n,$$

i.e. passing from G_1 to G_2 moves n elements from order k_1 to order k_2 and leaves every other order count unchanged.

In a single-bucket-swap, the gap in P_2 has the closed form

(1)
$$P_2(G_2) - P_2(G_1) \;=\; \frac{1}{|G|^2}\Big[\big(N_{k_2}(G_1)+n\big)^2 - N_{k_2}(G_1)^2 + \big(N_{k_1}(G_1)-n\big)^2 - N_{k_1}(G_1)^2\Big].$$

3. Main computational results

3.1. Distinct values on order 120.

Theorem 3. *There are* 47 *non-isomorphic groups of order* 120*. Their* 47 *values of* $P_2(G)$ *are pairwise distinct.*

Proof. A direct enumeration over `AllSmallGroups`(120) in GAP [3] computes each $N_k(G)$ from the conjugacy class table, then $P_2(G)$ as an exact rational. The 47 values, sorted ascending, are listed in Table 1. They are pairwise distinct. The reproducibility script is given in Section 5. □

Remark 4. Theorem 3 should not be over-interpreted: P_2 is *not* a complete invariant for finite groups in general. Already at order 16 the order-type alone fails to distinguish, since $\mathbb{Z}_4 \times \mathbb{Z}_4$ and $\mathbb{Z}_2 \times Q_8$ share the same multiset $\{1, 3, 12\}$ (one identity, three involutions, twelve elements of order 4), hence the same $P_2 = (1 + 9 + 144)/256 = 154/256 = 77/128$. So the smallest order where P_2 fails to separate non-isomorphic groups is ≤ 16, and Theorem 3 is a coincidence of order 120, not a structural fact.

3.2. **Single-bucket-swap pairs at order 120.**

Proposition 5. *Among the* $\binom{47}{2} = 1081$ *unordered pairs of groups of order* 120*, exactly* 27 *are single-bucket-swap pairs.*

Proof. By direct enumeration; see Table 2 and the GAP script. ☐

Example 6 (S_5 vs $\mathrm{SL}(2, 5)$)**.** The two well-known non-abelian non-solvable groups of order 120 that contain A_5 as a central or quotient factor are S_5 and $\mathrm{SL}(2, 5)$ (the binary icosahedral group, double cover of A_5). Their order-type distributions, computed from their conjugacy class tables, are:

$$\begin{aligned} S_5: &\quad N_1 = 1,\ N_2 = 25,\ N_3 = 20,\ N_4 = 30,\ N_5 = 24,\ N_6 = 20, \\ \mathrm{SL}(2,5): &\quad N_1 = 1,\ N_2 = 1,\ N_3 = 20,\ N_4 = 30,\ N_5 = 24,\ N_6 = 20,\ N_{10} = 24. \end{aligned}$$

The two distributions differ in exactly two orders: order 2 (where S_5 has 25 elements and $\mathrm{SL}(2, 5)$ has only the unique central involution) and order 10 (which has no representatives in S_5 but 24 in $\mathrm{SL}(2, 5)$). Passing from $\mathrm{SL}(2, 5)$ to S_5 redistributes 24 elements from order 10 to order 2. By (1),

$$P_2(S_5) - P_2(\mathrm{SL}(2,5)) = \frac{1}{120^2}\left[25^2 - 1^2 + 0^2 - 24^2\right] = \frac{624 - 576}{14400} = \frac{48}{14400} = \frac{1}{300}.$$

TABLE 1. The 47 groups of order 120 sorted by $P_2(G)$. "SG$(120, k)$" denotes the k-th entry in the GAP `SmallGroups` library at order 120.

k	Structure	P_2
17	$C_{12} \times D_{10}$	181 / 1440
4	C_{120}	187 / 1440
20	$C_3 \times ((C_{10} \times C_2) \rtimes C_2)$	43 / 288
5	$\mathrm{SL}(2, 5)$	1427 / 7200
34	S_5	1451 / 7200
35	$C_2 \times A_5$	1457 / 7200

3.3. **The full P_2 value table.**

TABLE 2. The 27 single-bucket-swap pairs at order 120, sorted by n (number of elements moved). Each row is realised by passing from group A to group B moving n elements from order k_1 to order k_2. The full table is in the reproducibility output.

A	B	n	k_1	k_2	$\lvert P_2(A) - P_2(B)\rvert$
5	34	24	10	2	1 / 300
5	35	30	4	2	1 / 240

4. Discussion

The result of Theorem 3 is a discrete coincidence: at order 120, the 47-element image of P_2 in $\mathbb{Q}$ happens to be injective. There is no reason to expect this on every order. Indeed, the order-type itself fails to be a complete invariant from order 16 onward, and any function of the order-type inherits this limitation in the worst case.

A natural question is to characterise the orders n at which P_2 is injective on the set of isomorphism classes. Computer experiments accessible to a workstation can answer this for $n \leq 256$ or so; we leave the systematic computation, and the question of structural reasons (if any) for injectivity at specific orders, to follow-up work.

The single-bucket-swap pattern of Proposition 5 is a rigidity condition: most pairs of groups of order 120 differ in three or more order-buckets. The 27 that differ in exactly two are a small minority and tend to occur between groups that are structurally close (sharing a normal subgroup, or being central extensions of the same quotient by isomorphic kernels). Whether "single-bucket-swap" admits an algebraic characterisation in terms of standard group-theoretic relations is open.

5. Reproducibility

All numerical claims are reproducible from the GAP script `reproducibility.g`, which we list in full below. On a workstation the entire computation runs in under one second.

```
LoadPackage("smallgrp");

n := 120;
num := NumberSmallGroups(n);
results := [];

for i in [1..num] do
    G := SmallGroup(n, i);
    desc := StructureDescription(G);
    ccs := ConjugacyClasses(G);
```

```
    orders := List(ccs, c -> Order(Representative(c)));
    sizes  := List(ccs, Size);
    distinct_orders := Set(orders);
    Nk := [];
    for k in distinct_orders do
        nk := Sum(Filtered([1..Length(orders)], j -> orders[j] = k),
                  j -> sizes[j]);
        Add(Nk, [k, nk]);
    od;
    p2 := Sum(Nk, x -> (x[2]/n)^2);
    Add(results, rec(id := i, desc := desc, Nk := Nk, p2 := p2));
od;

# Theorem: distinct P_2
distinct_p2 := Set(List(results, r -> r.p2));
Print("Distinct P_2 count: ", Length(distinct_p2), "\n");

# Proposition: 27 single-bucket-swap pairs
swap := 0;
for i in [1..num] do for j in [(i+1)..num] do
    a := results[i].Nk;; b := results[j].Nk;;
    all_k := Union(List(a, x -> [x[1]]), List(b, x -> [x[1]]));
    diff := [];
    for k in all_k do
        na := First(a, x -> x[1] = k); if na = fail then na := [k, 0]; fi;
        nb := First(b, x -> x[1] = k); if nb = fail then nb := [k, 0]; fi;
        if na[2] <> nb[2] then Add(diff, [k, nb[2] - na[2]]); fi;
    od;
    if Length(diff) = 2 and diff[1][2] + diff[2][2] = 0 then
        swap := swap + 1;
    fi;
od; od;
Print("Single-bucket-swap pair count: ", swap, "\n");
```

A worked Python post-processing script (parsing the GAP output, sorting the P_2 values, and listing the 27 swap pairs) is included in the supplementary materials accompanying this preprint.

References

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